

Mobil™

Unit ID: 533.LQ145

Asset ID: 52305238

Description: Cement Mill #3 Gearbox Lubrication Unit

Account Information

Sample Information

Equipment Information

ID: 462031

Name: ASEC - ACC

Address: , , EG

Sample ID: 26124405249

Service Level: Enhanced

Bottle ID: b094006223

Tested Lubricant: MOBILGEAR 600 XP 460

Asset Class: Circulating System

Manufacturer: Flender

Model:

Lubricant: MOBILGEAR 600 XP 460

Sample Data & Trends

Report Status

Alert

Sample ID

26124405249

Service Level

Enhanced

Bottle ID

b094006223

Tested Lubricant

MOBILGEAR 600 XP 460

Sampled

27 Apr 2026

Reported

05 May 2026

Equipment Age

Oil Age

Make-up Volume

Oil Changed

Filter Changed

Viscosity

466.4

27 Apr 2026

Visc@40C (cSt)

Wear

400

27 Apr 2026

Ag (Silver)

Al (Aluminum)

Cr (Chromium)

Cu (Copper)

Fe (Iron)

Ni (Nickel)

Pb (Lead)

Sn (Tin)

Contaminants

750

27 Apr 2026

Na (Sodium)

K (Potassium)

Si (Silicon)

Physical Properties

6

27 Apr 2026

Water (Vol%)

Oxidation (Ab/cm)

Lubricant

Contamination Rating

Alert

Equipment Rating

Alert

Lubricant Rating

Normal

ISO Code (4/6/14)

Particle Count >4um

Particle Count >6um

Particle Count>14um

PQ Index

783

Visc@40C (cSt)

466.4

Oxidation (Ab/cm)

4

Water (Vol%)

0.300

Wear (ppm)

Ag (Silver)

0

Al (Aluminum)

165

Cr (Chromium)

2

Cu (Copper)

43

Fe (Iron)

327

Mo (Molybdenum)

0

Ni (Nickel)

0

Pb (Lead)

1

Sn (Tin)

18

Contaminant (ppm)

K (Potassium)

29

Na (Sodium)

37

Si (Silicon)

588

Additive (ppm)

B (Boron)

10

Ba (Barium)

0

Ca (Calcium)

2224

Mg (Magnesium)

0

P (Phosphorus)

387

Zn (Zinc)

18



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Lubricant: MOBILGEAR 600 XP 460

Recommendation/Comments

ACTION REQUIRED - ELEVATED WEAR METALS LEVELS. Determine cause of elevated wear metals levels and take corrective action. If the wear metals levels are elevated for the first time with no history of an increasing trend, then resample immediately to confirm condition. Because silicon levels are not also elevated, abrasive dirt is not the cause of this condition. Possible causes of elevated wear metals include: a. Worn or oil-wetted parts; b. Contamination from previous fill; c. Over-extended oil service life; d. Normal break-in of new equipment. Continue to closely monitor the equipment and oil for signs of further condition regressions. Contact your ExxonMobil representative for further assistance if necessary.

ACTION REQUIRED - ELEVATED IRON LEVEL. Determine source of Iron (Fe) and take corrective action. If the iron level is elevated for the first time with no history of an increasing trend, then resample immediately to confirm condition. Possible sources of iron include: a. Machinery component wear; b. Metal-to-metal contact; c. Corrosion; d. Abrasive dirt contamination. The oil may be unfit for continued use and should be closely monitored for further condition regressions. Inspect the equipment for signs of damage. Change the oil after corrective measures have been implemented. Contact your ExxonMobil representative for further assistance if necessary.

ACTION REQUIRED - EXCESSIVE FERROUS DEBRIS. Determine cause of ferrous debris and take corrective action. If the ferrous debris level is elevated for the first time with no history of an increasing trend, then resample immediately to confirm condition. Potential causes of excessive ferrous debris include: a. Abnormal metallic component wear caused by abrasive dirt (or other contaminants); b. Material fatigue from heavy loads or other adverse operating conditions; c. Corrosive wear due to water, acids, or other contaminants in the lubricant; d. Normal "break-in" wear of new or recently replaced equipment. Inspect the equipment for signs of damage. Change the oil after corrective measures have been implemented. Contact your ExxonMobil representative for further assistance if necessary.

ACTION REQUIRED - ELEVATED SILICON LEVEL. Determine source of Silicon (Si) and take corrective action. If the silicon level is elevated for the first time with no history of an increasing trend, then resample immediately to confirm condition. If the wear metals levels are low, then the silicon likely exists in a non-abrasive form. Possible sources of non-abrasive silicon include: a. Defoamants; b. Sealant compounds. If wear metals levels are elevated, then the silicon likely exists in an abrasive form. Possible sources of abrasive silicon include: a. Abrasive dirt; b. Contamination from environment during an equipment overhaul or shutdown. Other possible sources of silicon include: a. Inefficient or leaking air intake filters; b. Leaking pipes; c. Clogged oil bath air filter; d. Poor oil filtration; e. Dirty sample containers or improper sampling procedures. In landfill gas applications, silicon can enter the engine as a siloxane. This material is a gas and is not normally abrasive; however, siloxanes can easily mask the presence dirt. The oil may be unfit for continued use and should be monitored closely for further condition regressions. Prolonged usage of lubricants with elevated levels of abrasive silicon can cause severe wear and damage to the equipment. Inspect the equipment for signs of excessive wear and change the oil immediately if damage is detected. Contact your ExxonMobil representative for further assistance if necessary.

ACTION REQUIRED - EXCESSIVE WATER CONTAMINATION. Determine source of water and take corrective action. Analyze the oil with an on-site water test if accessible. Potential sources of water include: a. Leaking seals; b. Condensation resulting from low operating temperatures or prolonged shutdowns; c. Compromised oil cooler; d. Direct exposure to atmosphere as a result of compromises to system components (i.e. improperly sealed tank fill access point. Remove water from oil using centrifuges or dehydration units, or drain the oil and recharge the system with new oil. Resample the system after corrective measures have been implemented. Contact your ExxonMobil representative for further assistance if necessary.

ACTION REQUIRED - UNSATISFACTORY OIL CONDITION. Test results indicate that the condition of this oil sample is unsatisfactory. The oil may be unfit for continued use and should be closely monitored further condition regressions. Contact your ExxonMobil representative for further assistance if necessary.

ACTION REQUIRED - EXCESSIVE WEAR METALS LEVELS. Determine cause of excessive wear metals levels and take corrective action. Possible causes of excessive wear metals levels include: a. Excessively worn bearings, gears, and / or shafts; b. Equipment fatigue; c. Abrasive dirt. The components suffering from excessive wear are unfit for continued use and should be inspected immediately. Change the oil immediately after corrective actions have been implemented. Contact your ExxonMobil representative for further assistance if necessary.

		Alert	
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Account Information		Sample Information	Equipment Information
ID: 462031 Name: ASEC - ACC Address: ,, EG		Sample ID: 26124405249 Service Level: Enhanced Bottle ID: b094006223 Tested Lubricant: MOBILGEAR 600 XP 460	Asset Class: Circulating System Manufacturer: Flender Model: Lubricant: MOBILGEAR 600 XP 460
ADMINISTRATION - PARTICLE COUNT DATA NOT REPORTED. The sample is heavily contaminated with visible sediment. Oil samples that contain significant levels of any contaminant can damage the laboratory's precision equipment. As a result, the particle count test was not completed, and analysis could not be provided. Before sending future samples to the laboratory, please ensure that no signs of visible sediment contamination are present. Consider the following: 1. The sample may not be representative of the system if it was acquired from a non-turbulent area (i.e. tank bottom); 2. Installing sample ports helps ensure that the sample is taken from the same location every time; 3. Tank fill caps and breathers might not be properly secured; 4. The filter / centrifuge may be operating incorrectly. If the sample was representative of the system, change the oil immediately to prevent equipment damage. Contact your ExxonMobil representative for further assistance if necessary.			
Sample Timeline			
27 Apr 2026 12:35 PM - Abd El Rahman Khaled - In Service Oil Sample: Comments: Photo:			